Mst Afsana Mimi

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Summary

Statistics master's graduate from UMass Amherst with experience in data pipelines and machine learning workflows on large datasets. Skilled in Python and SQL, with a strong background in data preprocessing and machine learning modeling.

Education

University of Massachusetts Amherst
Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

University of Dhaka
Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

University of Dhaka
Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Technical Skills

Programming & Querying

- Python (Pandas, NumPy, Scikit-learn)
- SQL (joins, aggregations, window functions)

Data Engineering & ML Workflows

- Data pipelines, ETL/ELT workflows, data preprocessing
- Data validation and quality checks
- Feature engineering and dataset construction

Tools & Platforms

- Git/GitHub
- Tableau
- AWS (S3, EC2) – familiarity

Statistical & ML Methods

- Regression, hypothesis testing, statistical modeling
- Machine learning model development and evaluation

Experience

Research Assistant
UMass Amherst Sep 21, 2026 – Present

- Data Systems Group, DREAM Lab, Manning College of Information and Computer Sciences.
- Project: Relational Predictive Query Processing. Supervisor: Marco Serafini.

Adjunct Faculty
UMass Amherst Sep 2026 – Present

- Department of Psychological and Brain Sciences.

Teaching Assistant (Data & Statistics)
UMass Amherst Sep 2023 – May 2026

- Supported student projects involving R/Python-based data analysis.

- Guided interpretation of statistical results and data-driven conclusions.

Research Assistant (NLP & Data Analysis)
University of Dhaka Jul 2021 – Jun 2022

- Processed large-scale text datasets and built NLP pipelines for data ingestion and transformation
- Developed and evaluated machine learning models using Python
- Implemented data preprocessing, feature engineering, and performance evaluation workflows
- Implemented data cleaning, validation, and consistency checks to ensure high-quality analytical datasets

Research Assistant (Socio-economic Data Analysis)
University of Dhaka Jul 2018 – Jun 2019

- Cleaned, transformed, and structured large-scale survey datasets for analysis.
- Built reproducible data processing workflows for statistical analysis.
- Applied statistical models to identify key socio-economic trends.
- Generated insights to support data-driven policy analysis.
- Maintained code and analysis workflows using Git for version control and collaboration.

Projects

Predictive Query over Relational Databases.

- Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data.
- Evaluating relational prediction workflows and developing a future benchmark for testing agentic systems on realistic database tasks.

Gaussian Process Modeling with Fairness Constraints

- Developing Gaussian Process models with fairness constraints across demographic groups.
- Designing evaluation strategies to assess trade-offs between predictive accuracy and bias.

High-Dimensional Feature Selection (MCAR)

- Designed feature selection methods for high-dimensional data settings.
- Improved feature selection effectiveness in high-dimensional datasets.

Awards and Scholarship

- Conference Awards for Scientists (CASc) by International Society for Clinical Biostatistics (ISCB).
- National Science and Technology (NST) Fellowship Award by Education Board of Bangladesh.
- Dhaka University Merit Scholarship for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra